

Embedded controller

Renesas Electronics has unveiled an innovative energy-harvesting embedded controller that can eliminate the need to



use or replace batteries in IoT devices. Developed based on

Renesas' breakthrough SOTB (silicon-on-thin-buried-oxide) process technology, the embedded controller achieves extreme reduction in both active and standby current consumption.

The extreme low current levels of the SOTB-based embedded controller enables system manufacturers to eliminate the need for batteries through harvesting ambient energy sources such as light, vibration and flow.

Renesas Electronics Corp.
www.renesas.com

LoRa system-in-package

LoRa (Long Range) technology is extending the reach of the IoT by combining long-range wireless connectivity



with low-power performance. To accelerate the development of LoRa-based connected solutions, Microchip Technology

has introduced a highly-integrated LoRa system-in-package (SiP) family with an ultra-low-power 32-bit microcontroller (MCU), sub-GHz RF LoRa transceiver and software stack.

SAM R34/35 SiPs come with certified reference designs and interoperability with major LoRaWAN gateway and network providers, significantly simplifying the development process.

Microchip Technology Inc.
www.microchip.com

Energy-harvesting BLE switch

ON Semiconductor is defining new levels of ultra-low power operation in the IoT with the introduction of its Blue-



tooth Low Energy (BLE) switch reference design that operates entirely from harvested energy. The platform demonstrates

how RSL10 SiP can enable Bluetooth 5 devices that are battery-less and entirely self-powered, requiring no additional energy source.

ON Semiconductor
www.onsemi.com

MISCELLANEOUS

Ethernet switches

HARTING has developed a new Ethernet switch that enables individual-



ly-tailored plug-and-play network solutions for data-intensive applications. HARTING Ha-VIS eCon unmanaged 16-port gigabit switch offers the necessary

performance to cover the demands of processing several high-quality images and streams in parallel throughout the network.

HARTING India Pvt Ltd
www.harting.co.in

BLE module

Insight SiP has launched ISP1807-LR module, which it believes to be the first fully Bluetooth 5.0 Long Range-compliant RF module in a miniature SiP package. It measures only 8mm x 8mm x 1mm, making it the smallest of its type. The miniature Bluetooth Low Energy (BLE) module offers Bluetooth 5 stack including long range, high throughput, advertising extensions and improved coexistence along with IPv6 connectivity and mesh capabilities.

Insight SiP
www.insightsip.com

Biometric attendance device

Matrix Aadhaar-enabled biometric attendance device, COSEC VEGA FAXQ enables employees to directly mark their attendance, without having to



manually entering their Aadhaar number. They can either use a smart card or keypad on the device followed by their fingerprint authentication to mark their attendance.

Matrix Comsec Pvt Ltd
www.matrixcomsec.com

System-on-chip

MediaTek has launched Helio P70 system-on-chip (SoC), with an enhanced artificial intelligence (AI) engine combined with CPU and GPU upgrades for more powerful AI processing. Helio P70 comes with upgraded imaging and camera support, a gaming performance boost and advanced connectivity features.

MediaTek Inc.
www.mediatek.com



Electronics project kits

Mega-Kit has introduced DIY project kits for juniors and hobbyists. The kits are based on basic science concepts such as electricity and electromagnetism. Screw driver technology has been used, so no additional tools, such as solder gun, are required.

Some projects are magnetic flashing LED, salt cell, dancing doll, jumping frog, DC motor, DC generator, windmill, electro-mechanical lock or plunger and solar boat. The kits include all necessary parts with detailed instruction manuals.

Mega-Kit
www.mega-kit.com



Multi-functional panels

The compact TFT colour display EA uniTFT050 was especially developed for use in compact medical devices. Thanks to integrated command set and object-based layout software, modern user interfaces can be created conveniently via drag-and-drop, without expert knowledge.

uniTFT050's 12.7cm (5-inch) screen offers a vector graphic display of 800 x 480 pixels. The extremely-fast image buildup of 50fps guarantees a streak-free display even with moving objects and animations. Extraordinary brightness of 900cd/m² guarantees a high-contrast display under all lighting conditions, including direct sunlight.

Electronic Assembly GmbH
www.lcd-module.com

